🌿 4Algae Liquid Tree Dashboard — Explained Simply

> A plain-English walkthrough of every section, model, and feature on the dashboard — written for someone with no prior background.

🧭 What is this project overall?

Think of this as a **smart monitoring + AI control room for a glass tank full of green algae** (called *Chlorella vulgaris*) placed in urban spaces like street poles or buildings.

The algae:

- **Eats CO_2** from the air (pollution) and releases **oxygen**
- Grows in water inside a transparent column called a **Photobioreactor (PBR)**
- Needs monitoring (temperature, light, pH, etc.) to stay healthy

The dashboard uses **AI + software simulations** to monitor, predict, and control this algae culture — like a pilot's cockpit, but for a living green machine.

📦 PART 1 — Top Status Bar (Navigation)

What it shows:

- **"DEMO STATUS: NOMINAL"** — the system is running fine
- **"DATA SOURCE: SIMULATED URBAN PBR"** — the sensor numbers are simulated (generated by software), not from real physical sensors

What it means:

> No real hardware is connected. Everything is a realistic simulation to demonstrate what the system *would* look like in a real deployment.

📊 PART 2 — Overview Stat Cards (Top Row — 4 Big Cards)

These are the **at-a-glance health numbers** for the algae reactor.

🟢 Card 1: Culture Density

Term	Simple Meaning
Culture Density	How many algae cells are packed in each drop of water
×10^6 cells/mL	Millions of cells per millilitre — like counting people per km^2
Est. Dry Biomass	If you dried the algae completely, how much solid material is left (in grams per litre)

> ⚠️ This number is **not measured** — it's labelled "OD680 EQUIV" (optical density equivalent), meaning we're using a known scientific formula to *estimate* it, not physically measure it. Real measurement requires lab equipment.

🟢 Card 2: Illustrative CO_2 Estimate

Term	Simple Meaning
CO_2 / day	Estimated kilograms of carbon dioxide the algae absorbs per day
600 L Working Vol.	The tank holds 600 litres of algae water
"Illustrative only"	This is a calculated estimate, not a gas meter reading

> ⚠️ The CO_2 number comes from a **formula** (`Biomass × Carbon fraction × 44/12`). It's not from a sensor — so it's shown as "illustrative" to be scientifically honest.

🟢 Card 3: Culture Lifecycle

Term	Simple Meaning
Day 8 — Batch Cycle	The algae has been growing for 8 days in this tank cycle
Exponential Phase	The algae is growing fastest right now (like teenagers — rapid growth phase)
Harvest Readiness: 76.7%	The tank is ~77% ready to be harvested (some culture drained, fresh nutrients added)

> 🌱 Algae growth goes through 4 phases: Lag → **Exponential** → Stationary → Decline. We're in the fast-growth exponential stage.

🟢 Card 4: Reactor Profile

Term	Simple Meaning
C. vulgaris	Short for *Chlorella vulgaris* — the type of algae used
Phototrophic	It gets energy from light (like plants)
LT-01	Reactor ID: "Liquid Tree Unit 01"
16h:8h Diurnal	Lights on 16 hours, off 8 hours — mimics a natural day/night cycle
"No Hardware Control"	Confirms: no real sensors or valves are connected; it's all simulated

🌡️ PART 3 — Bioreactor Environmental Telemetry (Sensor Panel)

This panel shows **6 simulated sensor readings** in real-time — the "vital signs" of the algae tank.

Each sensor card shows:

- **Current value** (big number)
- **A coloured bar** showing where it sits in the safe range
- **Min / Optimal / Max** thresholds

Sensor	What it measures	Optimal Range	Why it matters
Culture pH	Acidity/alkalinity of the water	7.0 – 7.8	Too acidic or alkaline kills algae
Temperature (°C)	Water temperature	20 – 26°C	Too hot/cold slows growth or kills cells
Dissolved O_2 (mg/L)	Oxygen dissolved in water	6 – 10 mg/L	Algae produce O_2; too much is toxic
Inlet CO_2 (ppm)	CO_2 being fed into the water	800 – 2000 ppm	Algae eat CO_2 to grow
Light Intensity (PAR)	Photosynthetically Active Radiation	150 – 300 µmol	The "food energy" for photosynthesis
Turbidity (NTU)	How cloudy/dense the water is	Low = sparse culture	High turbidity = dense algae = more biomass

> 🔵 These values are **simulated** by Python code with realistic random variation — they aren't from real sensors.

🤖 PART 4 — Next-Day Growth Predictor (AI Model 1)

What it does:

> "Given today's conditions, how many algae cells will there be **tomorrow**?"

The Model Used: **Random Forest Regressor** (scikit-learn)

- A **Random Forest** is like asking 100 different "decision trees" (flowcharts) for their prediction, then averaging all the answers
- It's one of the most reliable machine learning models for tabular (spreadsheet-style) data
- **Trained on**: Real Photobioreactor growth datasets from Zenodo (open science repository) — includes ~3,000 real experiment data points from lab PBRs

Inputs it uses:

Input Feature	What it is
Current cell density	How dense the culture is today
Temperature	Water temp
pH	Acidity level
Light intensity (PAR)	How much light is hitting the water
Culture age (day)	Which day of the batch cycle

What it outputs:

- **Predicted density tomorrow** ($\times 10^6$ cells/mL)
- **Delta** (change from today)
- **Confidence interval** (range of likely values)

⚠️ Important honesty label:

The model uses **Leave-One-Experiment-Out (LOEO) cross-validation** — meaning it's tested on entirely new experiments it never saw during training. The real-world accuracy ($R^2 = 0.726$) is significantly lower than the naive number ($R^2 = 0.928$), revealing how hard it is to generalise across different culture batches.

☠️ PART 5 — Culture Crash Risk Detector (Novel Feature 1)

What it does:

> "Is the algae culture about to die or collapse?"

No single sensor tells you this — so we invented a 5-factor Biophysical Stress Index:

Factor	What it checks	Why
Self-Shading Score	If the culture is *too* dense, cells in the middle get no light	Light can't penetrate thick green water
pH Drift Score	Is the pH moving away from the optimal zone fast?	Fast pH drift = CO_2 starvation or microbial contamination
Carbon Starvation Score	Is CO_2 supply low relative to cell density?	Algae starve without CO_2
Irradiance Stress	Is the light too high? Causes photoinhibition	Like sunburn for algae
Batch Age Penalty	Old cultures accumulate waste and decline	After day 12–14, cultures tire out

Each factor scores 0–1, then they're combined into a single **Crash Risk Score** (0–100%).

Output:

- 🟢 **LOW** — culture is healthy
- 🟡 **MEDIUM** — watch carefully, one factor elevated
- 🔴 **HIGH** — intervention needed immediately

> This is a **novel contribution** — most monitoring systems watch single sensors; this combines 5 biophysical signals into one early-warning score.

🛸 PART 6 — Biophysical Grey-Box Kinetics Engine (Novel Feature 2)

What it does:

> Uses physics equations + AI residuals together to model growth more accurately than either alone.

Two equations powering it:

1. Monod Substrate Saturation

- Classic microbiology formula: how fast algae grow depends on how much CO_2 is available
- Formula: `μ = μ_max × (S / (Ks + S))`
  - `μ` = growth rate
  - `S` = substrate (CO_2) concentration
  - `Ks` = half-saturation constant (how much CO_2 is "enough")
- **Simple version**: growth speeds up as CO_2 increases, but levels off at high concentrations (like eating — you speed up when hungry, but full = no faster)

2. Beer-Lambert Light Attenuation

- Light weakens as it travels through dense algae water
- Formula: `I(z) = I₀ × e^(-k × X × z)`
  - `I₀` = surface light intensity
  - `k` = extinction coefficient (how opaque the water is)
  - `X` = cell density
  - `z` = depth into the reactor
- **Simple version**: the deeper you go in a green tank, the darker it gets

The "Grey-Box" part:

- **White box** = pure physics (Monod + Beer-Lambert)
- **Black box** = pure AI (Random Forest)
- **Grey box** = both together — physics provides the structure, AI corrects the residual errors
- The AI learns "where the physics formula was wrong" and adjusts it

Outputs shown:

Output	Meaning
μ_net (h^{-1})	Net growth rate per hour
Core Photon Flux	Light actually reaching the centre of the tank
Monod Efficiency	What % of maximum possible growth rate is being achieved
Residual	How far off the physics model was (corrected by AI)

🔬 PART 7 — Microscopy Vision Panel (AI Model 2 — YOLOv8)

What it does:

> Automatically **counts individual algae cells** from microscope images — the same way a lab technician would do it with a hemocytometer, but using AI.

The Model: YOLOv8 Nano (Object Detection)

- YOLO = "You Only Look Once" — a state-of-the-art real-time object detection neural network
- **Nano** version = smallest, fastest variant (runs on CPU)
- **Custom trained** from scratch on microscopy images:
  - **Class 0:** `chlorella_cell` — a single algae cell (green dot)
  - **Class 1:** `cluster_noise` — groups of cells stuck together (excluded from count)
- **Trained for 20 epochs** (20 full passes through the training data)
- **Dataset**: CIDACC microscopy image dataset, prepared as a YOLO-format dataset

Why not use simpler methods?

> Classical methods (Otsu thresholding, Laplacian-of-Gaussian blob detection) were tested and completely failed — F_1 score = **0.012** — because the hemocytometer's **etched grid lines** look exactly like cell edges to classical detectors. YOLOv8 learned to ignore the grid.

What it shows on screen:

- 🟢 **Green bounding boxes** = individual Chlorella cells detected
- 🔴 **Red bounding boxes** = clusters/debris (excluded from count)
- **Big cell count number** = total cells in this microscope field of view

📐 PART 8 — Cellular Biovolume Estimator (Novel Feature 3)

What it does:

> Turns the 2D bounding boxes from YOLO into **3D physical volume estimates** of the algae biomass.

The Science: Prolate Spheroid Geometry

- Chlorella cells are **slightly oval** (prolate spheroid = stretched sphere, like a rugby ball)
- YOLO gives us a bounding box (width × height in pixels)
- We convert pixels → micrometres using a known scale (0.3125 µm/pixel)
- Then calculate volume using the prolate spheroid formula:
  - `V = (4/3) × π × a × b²`
  - `a` = semi-major axis (long side / 2)
  - `b` = semi-minor axis (short side / 2)

What it shows:

Metric	Meaning
Total Biovolume (µm³/FOV)	Total physical volume of all detected cells in the image
Mean Cell Diameter (µm)	Average cell size
Mean Cell Volume (µm³)	Average volume per cell
Est. Dry Mass (pg)	Estimated picograms of dry biomass (using 27% dry weight factor)

> 🔬 This bridges the gap between vision (counting) and biochemistry (mass). No other open-source algae monitoring tool does this automatically from microscopy images.

🎯 PART 9 — Uncertainty & Optical Triage (Novel Feature 4)

What it does:

> Tells you **how confident the AI was** in each detection, and flags images that need a human to double-check.

The Concept: Active Learning Triage

- Every YOLO detection has a **confidence score** (0–100%)
- Detections below **65% confidence** are "borderline" — the AI is unsure
- These borderline cases are **flagged for human review** (this is called "Active Learning" — the AI asks for help where it's uncertain)

What it shows:

Metric	Meaning
Mean Confidence	Average certainty across all detections
Borderline Count	Number of detections the AI was unsure about
Debris Noise %	What % of detections are clusters/noise (Class 1)
VERIFIED / BORDERLINE / FLAG	Traffic light for image quality

> If an image has many borderline detections, it means the microscope image is blurry, overexposed, or the culture is very clumped. A real system would send that image to a human annotator.

⚡ PART 10 — Cyber-Physical Actuator Hub (Automation Simulator 1)

What it does:

> Simulates the **closed-loop automatic control system** that would physically adjust the reactor conditions — like a thermostat, but for a living algae tank.

The Concept: Closed-Loop Control (AI-MPC)

- **Closed-loop** = the system reads a sensor → compares to target → adjusts something → reads sensor again → repeats
- **MPC** = Model Predictive Control — instead of just reacting (like a thermostat), it *predicts* what will happen and acts in advance

Three Actuators (Simulated):

🟢 CO_2 Solenoid Valve (SV-CO2-01)

- **What it does**: Opens a valve to inject CO_2 gas into the water
- **Why**: Algae need CO_2 to grow; pH drops when CO_2 increases
- **AI control logic**: If pH is rising above 7.6 (algae consuming CO_2 fast), valve fires a **micro-pulse** of CO_2
- **Simulated output**: Pulse duration (e.g., 2.3 seconds), target pH

🟡 Adaptive LED PWM Driver (LED-PWM-DIM-01)

- **What it does**: Adjusts the brightness of LED grow lights
- **Why**: Too much light = photoinhibition (algae get "sunburned"); too little = slow growth
- **AI control logic**: Uses Beer-Lambert equation to calculate how much light reaches the centre of the tank at current density, then dims/brightens LEDs to compensate
- **PWM** = Pulse Width Modulation — a way to control LED brightness electronically
- **Simulated output**: LED power % (e.g., 85%), PPFD (light intensity), core photon penetration %

🔵 Auto-Harvest Drain/Refill Valves (V-DRAIN-01 / V-REFILL-02)

- **What it does**: Drains some culture when density is too high, refills with fresh nutrients
- **Why**: Prevents self-shading and culture crash from overcrowding
- **AI control logic**: Triggers when density exceeds 24.0M cells/mL

- **Simulated output**: Volume to drain (litres), expected density after dilution

> 🚫 This is all **simulation** — no real valves are connected. But the code computes real physics-based outputs that a real control system would use.

🤖 PART 11 — Autonomous Mobile Liquid Tree (AMR Pod) — Automation Simulator 2

What it does:

> Simulates a **robot on wheels** carrying a small algae tank that **autonomously drives to wherever CO_2 is highest** inside a corporate building.

The Concept: Indoor Air Quality (IAQ) Dispatch

- A 60-litre Chlorella tank is mounted on an **Autonomous Mobile Robot (AMR)**
- The building has CO_2 sensors in each room (IoT mesh)
- When a room's CO_2 spikes (crowded meeting = everyone breathing heavily), the robot is dispatched there
- The algae scrubs CO_2 and releases O_2 in real-time

The Robot Specs (Simulated):

Spec	Value
Tank volume	60 L Chlorella culture
Drive system	Mecanum wheels (moves in any direction)
Navigation	2D LiDAR SLAM (maps the room, avoids obstacles)
Charging	Inductive wireless charger at base station
Harvest	Auto-drains to mother base when returning

Four Building Zones:

Zone	CO_2 Level	Priority
🔴 Executive Boardroom A	1,650 ppm	**URGENT** — Fatigue Risk
🟡 Conference Room B	1,050 ppm	Elevated
🟢 Engineering Open Desks	640 ppm	Normal
🔵 Autonomous Mother Base Dock	420 ppm	Base (ambient)

The AI Dispatch Logic (Monod Scrubbing Model):

When dispatched, the model calculates:

1. **Transit time** = distance ÷ robot speed
2. **Scrubbing rate** = using the **Monod equation** (same from Part 6)
    - How fast does 60L of algae absorb CO_2 from a 200m^3 room?
3. **Time to remediate** = how long to bring CO_2 from e.g. 1,650 → 750 ppm
4. **O_2 produced** = calculated from photosynthesis stoichiometry

What it shows:

Metric	Meaning
Target Zone	Where the robot is heading
Battery %	Simulated charge level
Transit ETA	Minutes to arrive
Est. Scrub Time	How long to fix the air quality
CO_2 Remediation	e.g., 1,650 → 750 ppm
O_2 Output	e.g., +88.5 g of oxygen generated

📈 PART 12 — 14-Day Historical Trends (Charts)

What it shows:

Four line charts showing how the key parameters changed over a **14-day batch cycle**:

Chart	What it tracks
Cell Density Trajectory	How algae population grew over 14 days (S-curve shape: slow → fast → plateau)
Daily CO_2 Uptake	How much CO_2 was absorbed each day (peaks in exponential phase)
Diurnal pH Modulation	pH rises during the day (photosynthesis consumes CO_2, raising pH) and falls at night
Water Temperature	Temperature variations over the cycle

> 📌 These are **hard-coded illustrative series** — not real experiment measurements. They show what a healthy 14-day batch cycle should look like.

📋 PART 13 — Operational Alerts & Recommendations

Alerts Panel:

- Shows **rule-based health flags** (e.g., "pH approaching upper limit", "culture age entering stationary phase")
- Each alert is coloured: 🟢 OK / 🟡 Warning / 🔴 Critical
- These are generated by checking sensor values against thresholds — no AI model, just `if-else` rules

Harvest & Sparging Advisory:

Recommendation	What it means
Culture Dilution / Partial Harvest	"Drain X litres and replace with fresh nutrients" — with % readiness
CO_2 Sparging Schedule	"Run CO_2 for 15 min, then off for 45 min" cycle
Photoperiod Schedule	"16 hours light, 8 hours dark"

> These are **experimental suggestions** — not validated instructions. They're based on general algae biology principles, not a certified control algorithm.

🧩 Summary — All the AI/Models at a Glance

Section	Model / Method	Type
Growth Prediction	**Random Forest Regressor** (scikit-learn)	Machine Learning
Culture Crash Risk	**5-Factor Biophysical Stress Index**	Custom Formula
Grey-Box Kinetics	**Monod + Beer-Lambert + RF Residuals**	Physics + ML
Cell Detection	**YOLOv8 Nano** (Ultralytics)	Deep Learning (CNN)
Biovolume Estimation	**Prolate Spheroid Geometry**	Computational Geometry
Uncertainty Triage	**Confidence Thresholding + Active Learning**	Statistical
CO_2 Valve Control	**AI-MPC Closed-Loop** (simulated)	Control Theory + ML
LED Dimming	**Beer-Lambert PWM** (simulated)	Physics-based Control
AMR Dispatch	**Monod IAQ Scrubbing Model** (simulated)	Physics Simulation

❗ Scientific Honesty Labels (What's Real vs. Simulated)

Label	Meaning
🔵 **SIMULATED**	Numbers generated by software code, not real sensors
🟡 **EXPERIMENTAL**	Based on real science, but not validated with physical hardware
🟢 **LIVE**	Would indicate real data (not used in current prototype)
⚠️ **Calibration Unavailable**	Density is estimated — not physically measured in this prototype

> 💡 **Bottom line**: This dashboard is a **fully functional research prototype** that demonstrates AI-driven algae monitoring. The science, models, and control algorithms are real and grounded in published research — the sensor data is simulated because no physical tank exists yet. In a real deployment, real sensors would replace the simulated values, and everything else would work the same way.